

EXPERIMENT TO VERIFY OHM'S LAW

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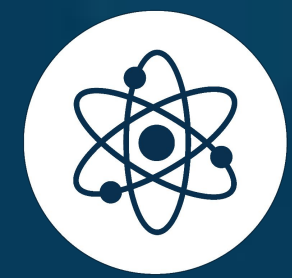
PhD Physics

University of Benin



AIM

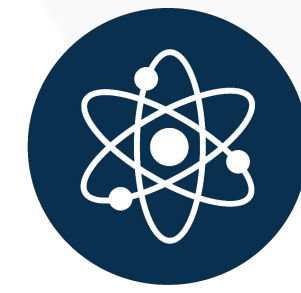
To verify Ohm's law.



LEARNING OBJECTIVES

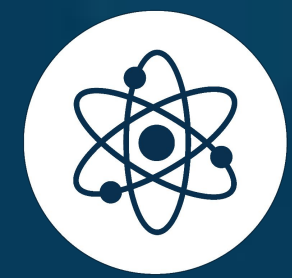
At the end of this lesson, you should be able to:

- 1. Analyse how current and voltage relate in a circuit with Ohmic resistance.**
- 2. Calculate the total resistance when resistors are connected in series.**
- 3. Calculate the total resistance when resistors are connected in parallel.**
- 4. Use Ohm's law and series/parallel resistance to solve circuit problems.**



APPARATUS

- **Breadboard**
- **Voltmeter and Ammeter**
- **DC power supply**
- **Connecting wires and Bridges (Jumpers)**
- **Two resistors R_1 and R_2**



THEORY

Ohm's law states that "The electric current, I passing through a conductor is directly proportional to the potential difference, V at constant temperature."

$$V = I R$$

where R is the resistance and is measured in Ohm (Ω).

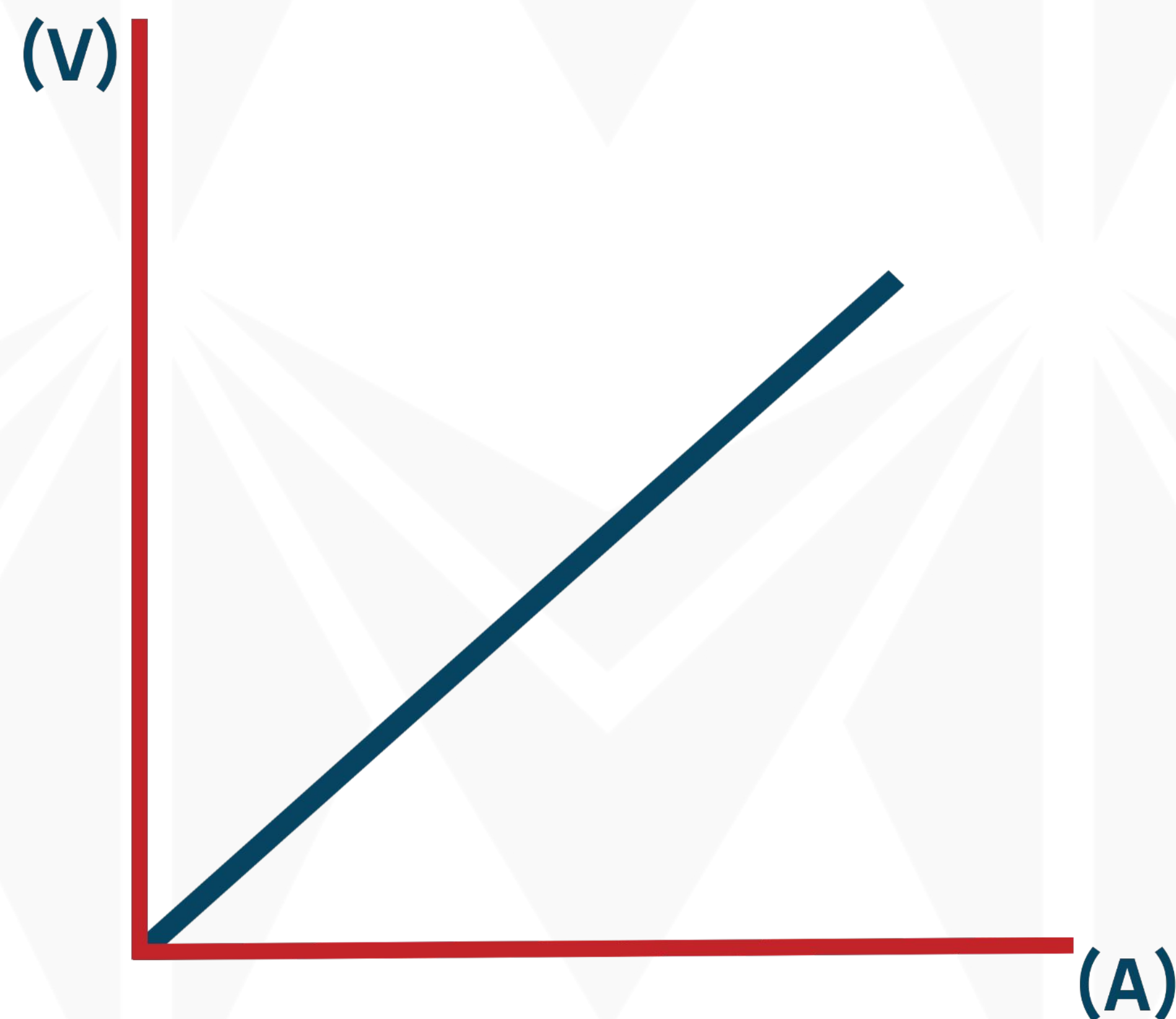
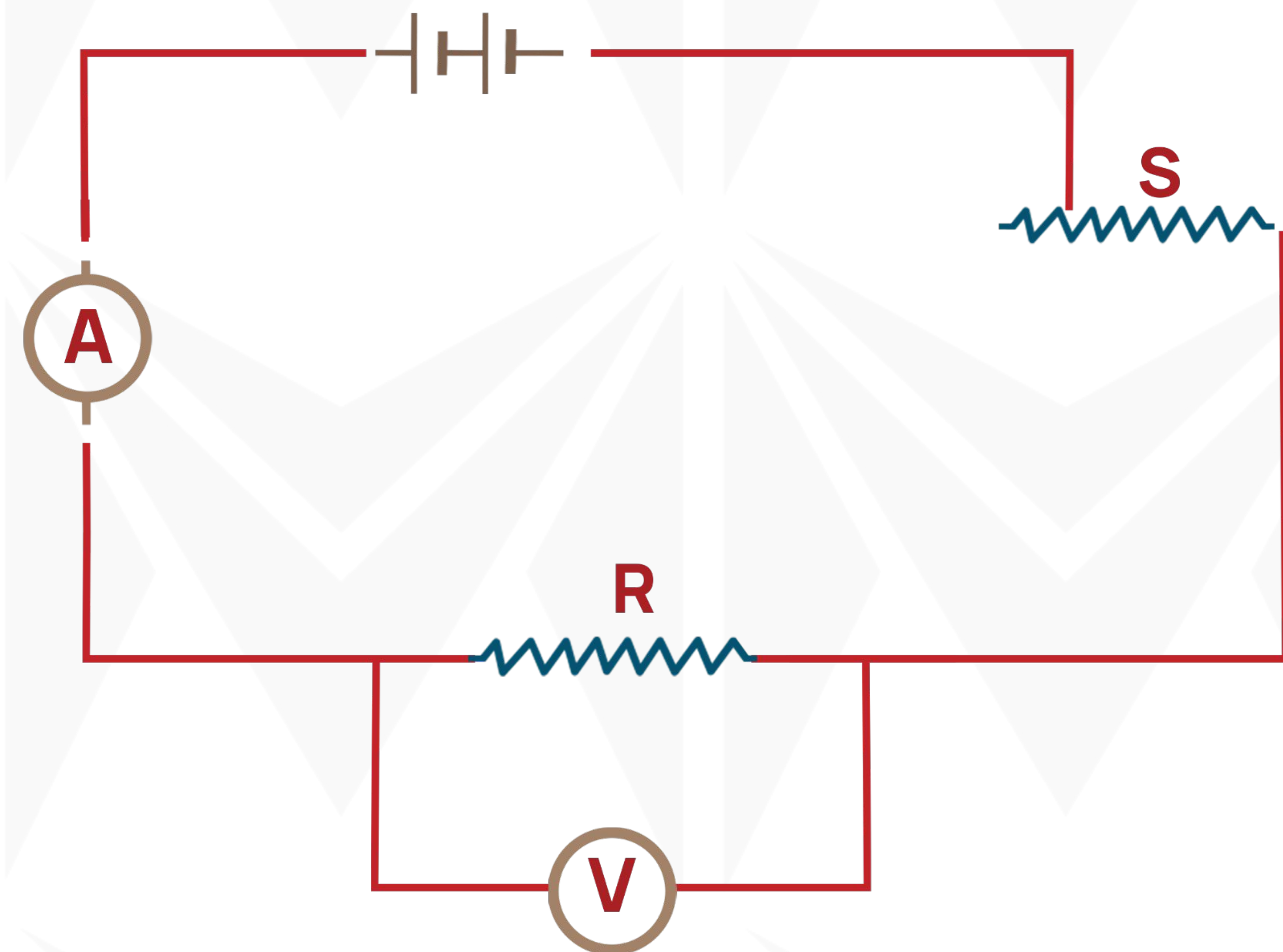
If we plot V versus I for Ohmic resistance, we obtain a straight line that passes through the origin whose slope is equal to the resistance, R

Ohm's Law was formulated by Georg Simon Ohm, a German physicist, in 1827.



THEORY

$V = I R$ compared to $y = mx + c$





THEORY

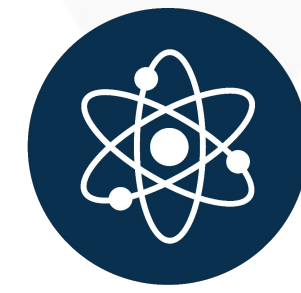
FACTORS AFFECTING THE VALUE OF THE RESISTANCE:

1. The length of the conductor: $R \propto l$
2. The cross-sectional area of the conductor: $R \propto \frac{1}{A}$
3. The type of the conductor's material: $R = \rho \frac{l}{A}$

where ρ is the resistivity of the conductor which depends on the type of the conductor's material and the temperature.

4. Temperature: $R = R_o [1 + \alpha (T - T_o)]$

where α is the temperature coefficient of resistance that depends on the type of the conductor's material and R_o is the resistance of the conductor at some reference T_o .



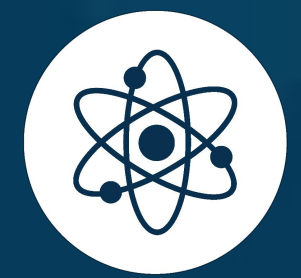
THEORY

What is a Resistor:

- A resistor is a device that resists the flow of current.
- The resistance of a resistor is measured in ohms (Ω).
- Resistors are used in electrical circuits to control the current and voltage.

Resistors can be connected in series and parallel





RESISTORS CONNECTIONS

Series Connection

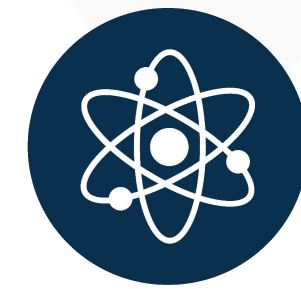
When the resistors are connected in series, in general the same current passes through them, while the potential difference is different (except when the resistors are equal).

The equivalent resistance, R of two (or more) resistors ($R_1, R_2, R_3, \dots, R_N$) connected in series is:

$$R = R_1 + R_2 \quad (\text{For two resistors in series})$$

$$R = R_1 + R_2 + R_3 + \dots + R_{N'} \quad (\text{For more than two resistors in series})$$

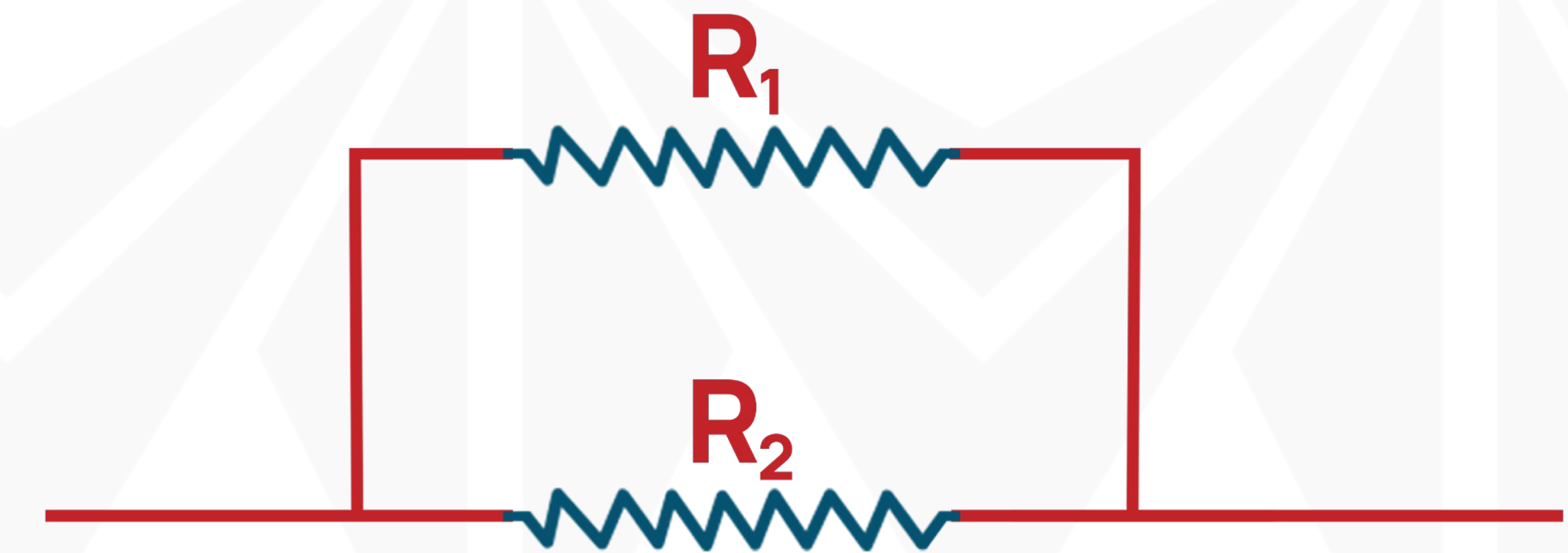




RESISTORS CONNECTIONS

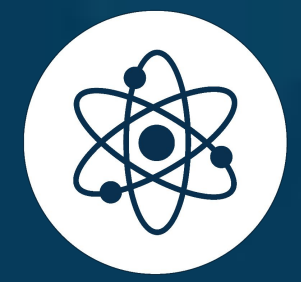
Parallel Connection

- When the resistors are connected in *parallel*, they all have the same *potential* difference, but the current that passes through any of them is different (except when the resistors are equal).
- The equivalent resistance, R of two resistors (R_1 and R_2) connected in *parallel* is:



- When 'more than 2 resistor' $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$ parallel:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$



MID-LESSON QUESTIONS

Question 1: What is the equivalent resistance in a **SERIES** circuit with three resistors ($R_1=10\Omega$, $R_2=20\Omega$ and $R_3=30\Omega$)?

Question 2: What is the equivalent resistance in a **PARALLEL** circuit with three resistors ($R_1=10\Omega$, $R_2=20\Omega$ and $R_3=30\Omega$)?



MID-LESSON ANSWERS

Answer 1: In a series circuit, the equivalent resistance:

$$R = R_1 + R_2 + R_3 = 10\Omega + 20\Omega + 30\Omega = 60\Omega$$

Answer 2: In a parallel circuit, the equivalent resistance:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

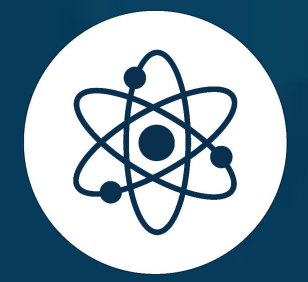
$$\frac{1}{R_{eq}} = \frac{1}{10\Omega} + \frac{1}{20\Omega} + \frac{1}{30\Omega}$$

$$\frac{1}{R_{eq}} = 0.1 + 0.05 + 0.0333$$

$$R_{eq} = \frac{1}{0.1+0.05+0.0333}$$

$$R_{eq} \approx \frac{1}{0.1833}$$

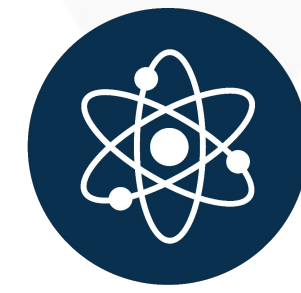
$$R_{eq} \approx 5.45\Omega$$



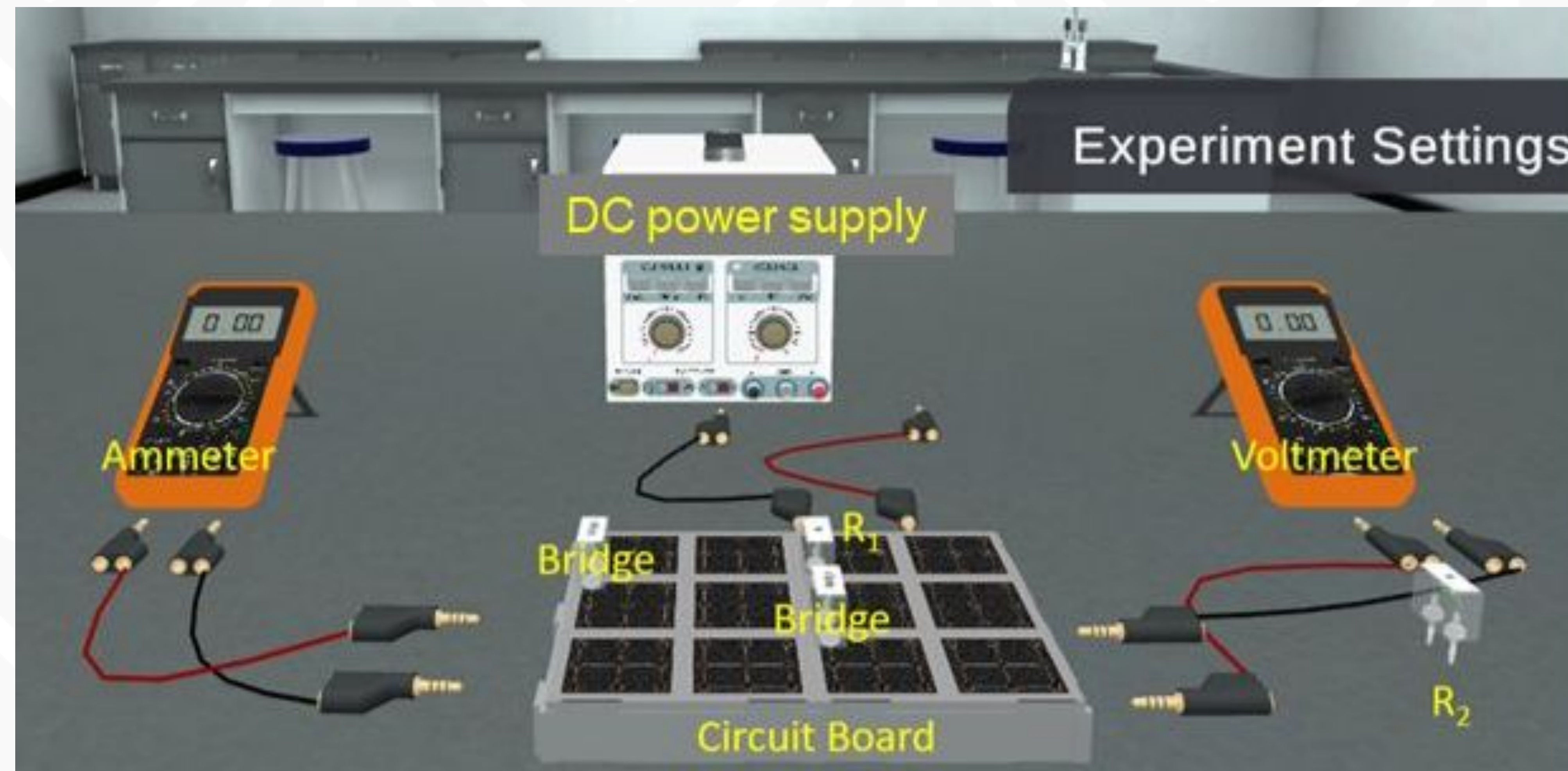
PROCEDURE

Start-up Screen of Ohmic Resistor Virtual Lab (Praxy labs)

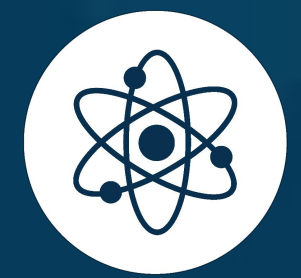




PROCEDURE

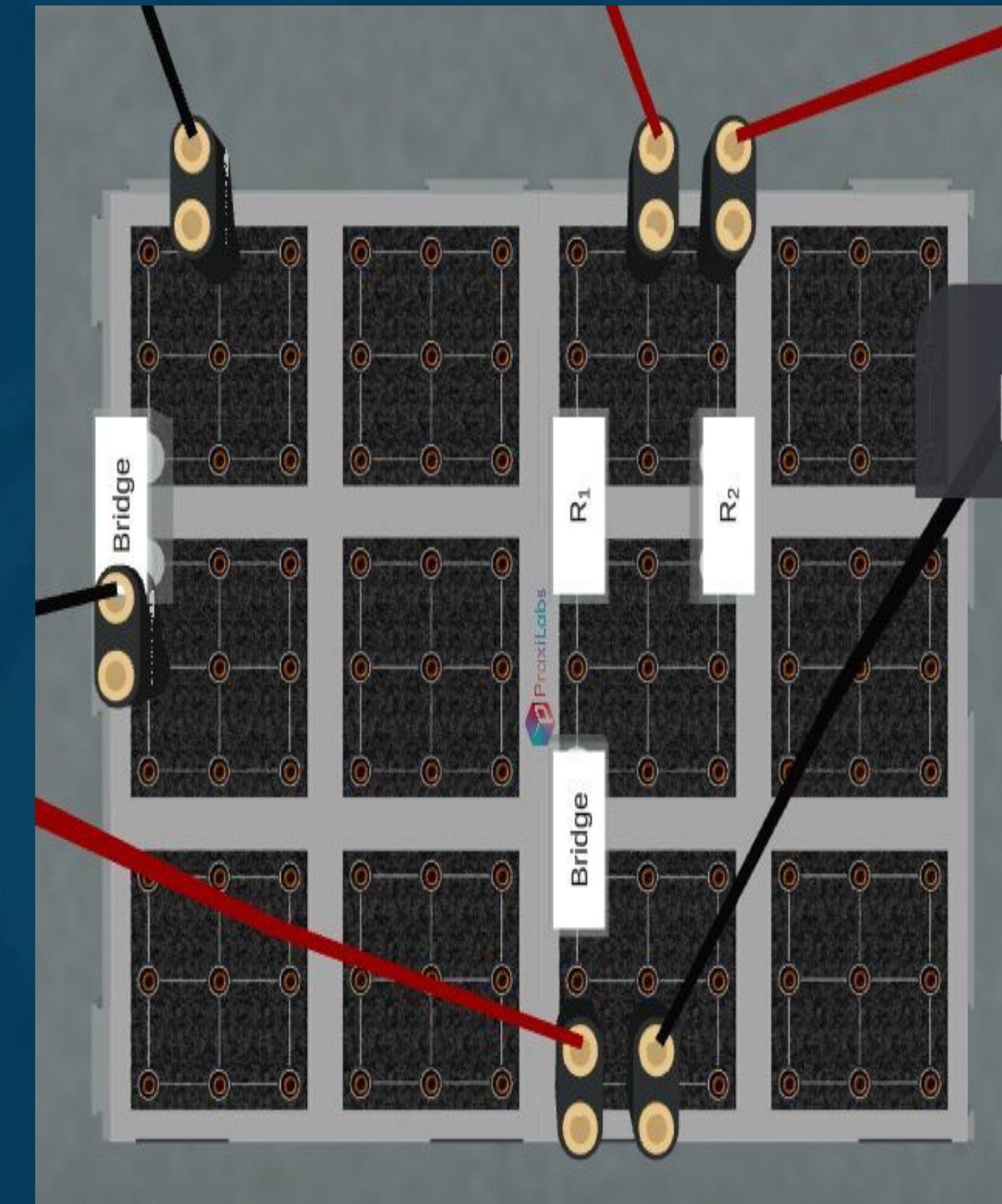


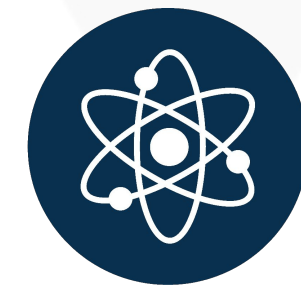
1. Click on the up arrow in the upper right corner of the UI to view the whole circuit clearly.
2. Using the scroll wheel of your mouse, zoom in on the circuit to identify its components.



PROCEDURE

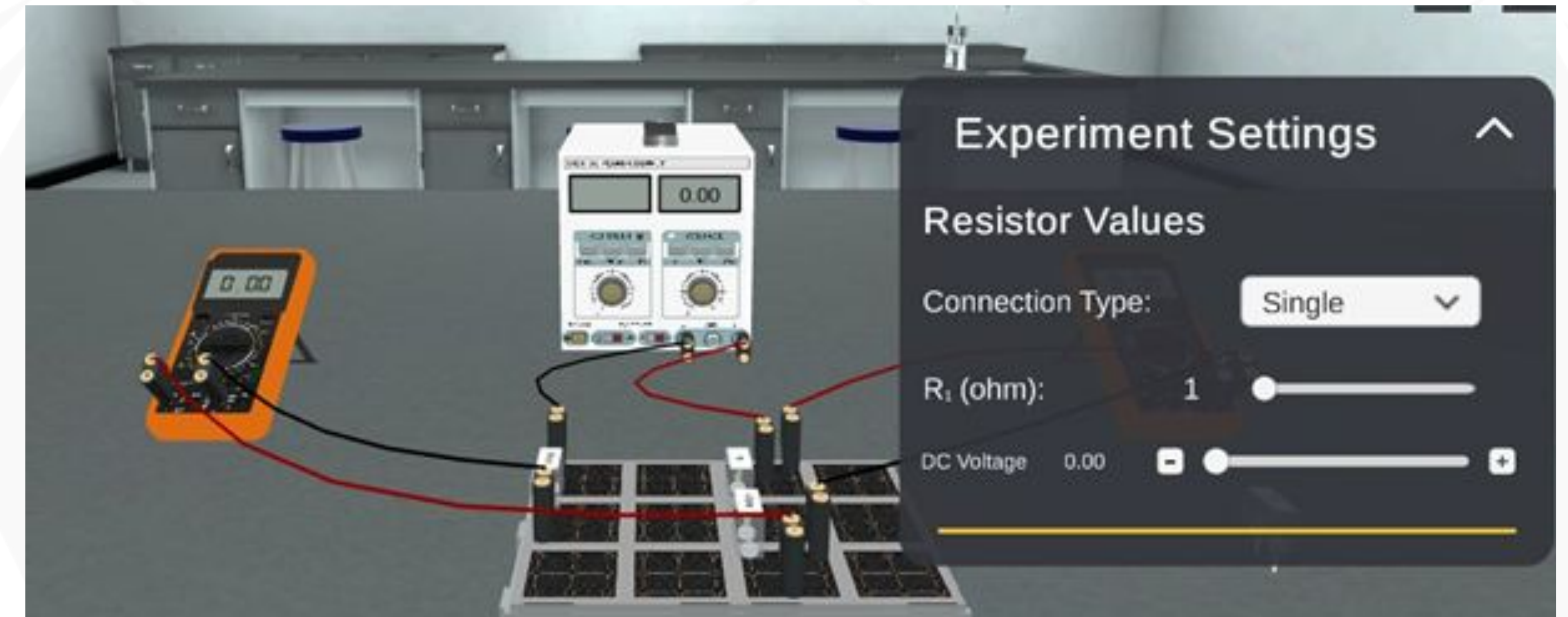
3. Click on the connection board to view the components and connect the circuit.
4. Connect the circuit as shown in the figure.
5. Click anywhere on the board to zoom out of the board.
6. Click on the voltmeter to zoom in and connect it as shown in the figure.
7. Click anywhere on the voltmeter to zoom out.

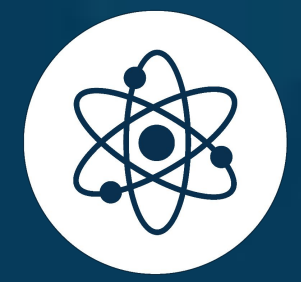




PROCEDURE

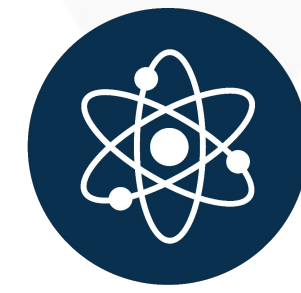
8. Click on the ammeter and connect it as shown in the figure.
9. Click anywhere on the ammeter to zoom out.
10. Click on the DC power supply and connect it as shown in the figure.
11. Click anywhere on the DC power supply to zoom out.
12. Expand the UI again by clicking on the down arrow.





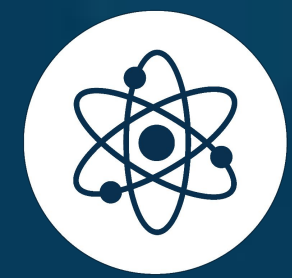
PROCEDURE

13. Set the value of $R1$ (1 – 20 $k\Omega$).
14. You can apply a potential difference to the resistance either by using the knob on the DC power supply or by using the DC voltage slider on the UI.
15. Adjust the voltage of the DC power supply to a certain value (say 1 V).



PROCEDURE

- 16. Click on the Record button to record both V and I .**
- 17. Repeat the last two steps six times for different values of the potential drop.**
- 18. Now the experiment has been ended, and an Excel sheet that contains the data you have recorded will be downloaded automatically**



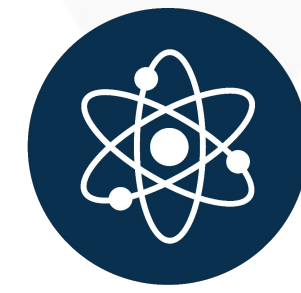
DATA ANALYSIS

1. The downloaded Excel sheet will look like that shown.
2. Plot a graph of $V(V)$ vs $I(mA)$, which yields a straight line that passes through the origin.
3. Deduce the slope of the line, which is equal to the magnitude of the resistance you have worked with (R_1 in this case).

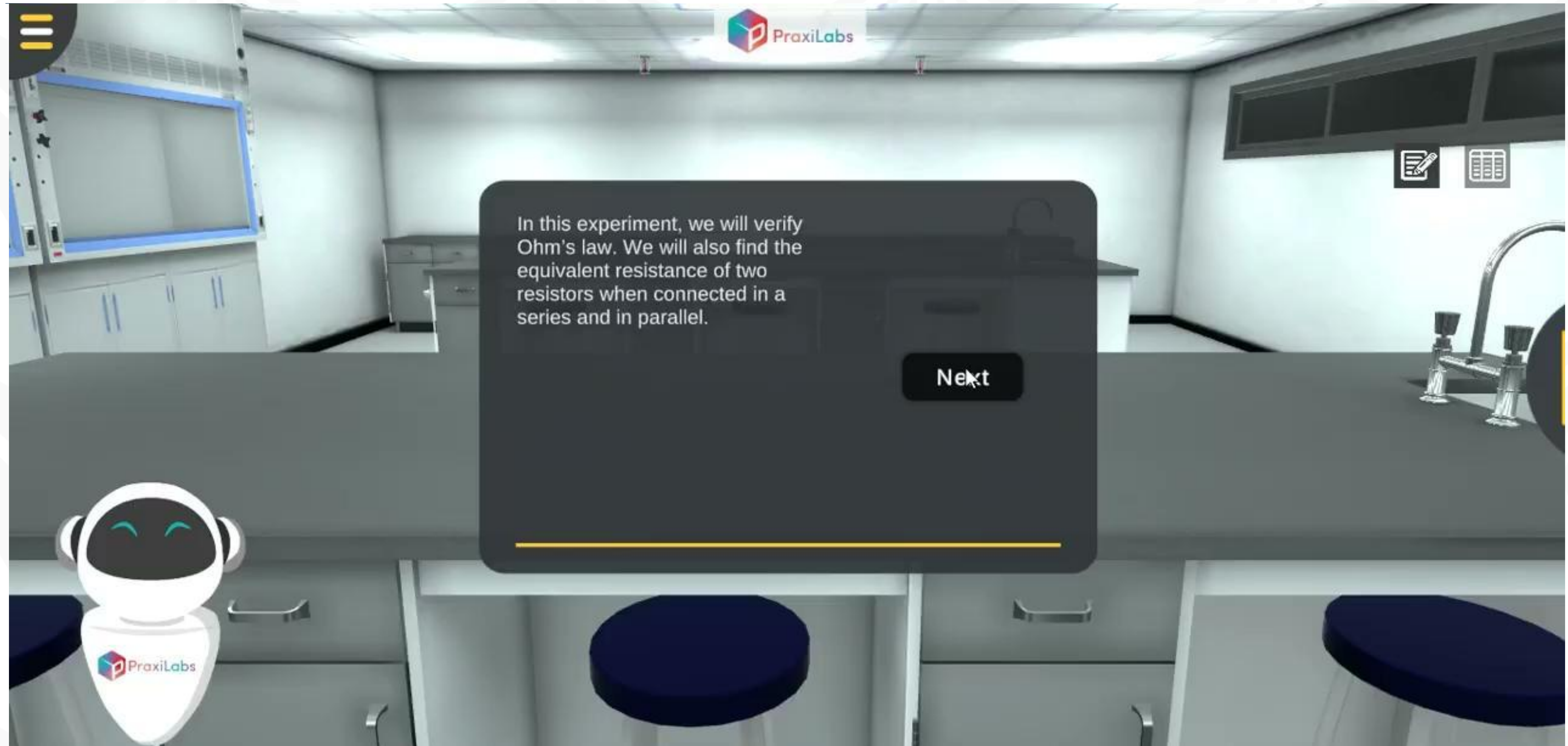
$$R_1 = \text{slope} = \frac{\Delta V}{\Delta I}$$

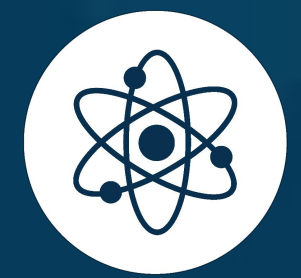
1. Using the value for the resistance R_1 , you can calculate the % error in the magnitude of R_1 (which should be within 5 % of its default value which 1 kiloOhms).

V (V)	I (mA)
0.1	0.1
0.4	0.4
0.5	0.5
0.7	0.7
0.9	0.9
1	1



WALK THROUGH VIDEO





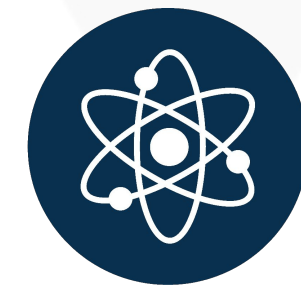
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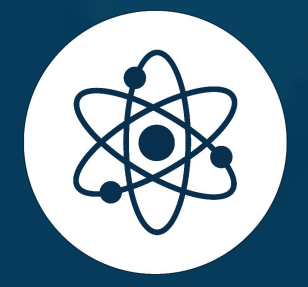
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<http://hyperphysics.phy-astr.gsu.edu>

<http://www.physicstutorials.org>



THANK
YOU